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ORIGINAL ARTICLE

SPORT INJURIES AND REHABILITATION

Alterations in hamstring properties of athletes with hamstring strain injuries can impact jump-landing performance

Amornthep JANKAEW [1], Cheng-Feng LIN [1, 2] *

[1]College of Medicine, Department of Physical Therapy, National Cheng Kung University, Tainan, Taiwan (ROC); [2]Physical Therapy Center, National Cheng Kung University Hospital, Tainan, Taiwan (ROC)

*Corresponding author: Cheng-Feng Lin, College of Medicine, Department of Physical Therapy, National Cheng Kung University, N. 1 University Road, 70101 Tainan, Taiwan (ROC). E-mail: connie@mail.ncku.edu.tw

ABSTRACT

BACKGROUND: Hamstring strains may alter the contractile properties of the muscle and affect functional movement. Thus, this study evaluated the association between the hamstring properties and the jump-landing performance in athletes with hamstring strains.
METHODS: Sixteen hamstring-injured athletes and 16 healthy controls were recruited. The mechanical properties and contractile function of the hamstring were assessed in both the injured and non-injured legs. The control group was tested only in the matched injured leg. The kinetic outcomes during squat jump (SJ) and countermovement jump (CMJ) tasks were analyzed to evaluate the jump-landing performance.
RESULTS: The injured limb exhibited higher muscle tone (P=0.042) and stiffness (P=0.010), but lower flexibility (P=0.002) and strength (P=0.040) than the control limb. The injured limb showed a poorer jumping performance (P=0.037 for jump height) and landing performance (P=0.011 for landing force and P=0.004 for loading rate) during the CMJ task compared to the control limb. All the hamstring properties showed mild-to-moderate correlations with the jump-landing performance.
CONCLUSIONS: Impairments in the muscle properties following hamstring strain impact the jumping and landing performance, leading to degraded sports outcomes in athletes with hamstring injuries. Thus, more attention should be paid to tissue property changes following hamstring strain injuries to develop effective strategies for restoring muscle function and improving sports performance.

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KEY WORDS: Athletes; Exercise; Muscles.

Hamstring strain injuries (HSI) are a common form of acute musculoskeletal injury in sports activities. The incidence of HSI is remarkably consistent over time, accounting for approximately 0.81 cases per 1,000 hours of exposure incurred during sprinting, jumping, and kicking.[1, 2] In addition, hamstring injuries carry a high risk of reinjury, ranging from 14% to 63%,[3] with recurrence occurring within the first two weeks following return to play in approximately one-third of the cases.[4, 5] Hamstring reinjuries often present with greater severity than the initial injury.[6] Consequently, HSI usually requires a long convalescence time, typically around 8-28 days or more, resulting in a prolonged absence from training and competition.[7] This not only affects athletic performance but may also have serious financial implications for team management.

Hamstring strains affect the morphology, mechanical properties, and function of the muscle[8] in both the acute[9] and chronic phases.[9, 10] Furthermore, post-HSI remodeling may lead to scar tissue formation in the affected region over a period of 6 weeks[11] to 23 months after injury.[12] This scar formation is thought to be a primary contributor to alterations in the muscle mechanical properties and function and results in various muscle impairments of the injured hamstring, including decreased strength and fascicle length,[13] reduced hamstring activation,[14] and changes in the reflex response[15] and motor performance.[16] Hamstring

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injuries may also prompt compensation in the coordination and function of the posterior chain muscles,[17, 18] which potentially limits the rehabilitation process, diminishes sports performance, and increases the risk of reinjury. Thus, a thorough assessment of the hamstring properties of HSI athletes is crucial in clinical rehabilitation to gather the information needed to formulate appropriate training programs and back-to-sports decisions. Notably, such assessments may also expedite the recovery process and assist in implementing preventive strategies capable of reducing the likelihood of future hamstring injuries.

The functions of the hamstrings are evidenced in various dynamic sports tasks.[19] Prior research has demonstrated that changes in the mechanical properties of the hamstring impact the jumping performance. For instance, Jankaew *et al.*[20] found that changes in the passive stiffness of the hamstrings affected the lower limb muscle recruitment in pain-free athletes, as observed in the electromyography power frequency spectrum, and had a direct impact on the jump-landing performance. Furthermore, Blackburn *et al.*[21] showed that increased hamstring stiffness resulted in reduced anterior tibial translation and an internal varus moment during single-leg and double-leg jump landings, which lowered the risk of anterior cruciate ligament injury. However, while these studies show a direct relationship between the hamstring properties and the recruitment levels during jump-landing, the precise effects of post-HSI changes in the hamstring mechanics on the performance of functional tasks such as jumping and landing are not fully understood. This lack of understanding is unfortunate given that jumping tasks form a crucial component of many training programs aimed at injury prevention and/or rehabilitation in different athletic populations.[22]

Accordingly, this study investigates the effects of the hamstring mechanical properties (muscle tone, stiffness, and elasticity) and contractile function (flexibility and strength) on the squat and countermovement jump performance of athletes with HSI and healthy individuals. It is hypothesized that these properties and functions are closely linked to the jump-landing performance. In particular, a poorer function of the hamstring after HSI is expected for the injured limb, leading to an impaired jump-landing performance compared to that of the non-injured limb or matched limb in healthy controls. The insights obtained regarding the effects of long-term muscle adaptations following HSI on functional activities such as jumping and landing provide a useful reference for the formulation of effective assessment and rehabilitation strategies.

TABLE I.—*Characteristics of HSI and control groups (mean±SD).*

Parameters	HSI	CON	*t*	P
N.	16	16		
Age (years)	21.94±2.02	21.06±1.34	1.337	0.191
Weight (kg)	67.30±6.00	63.19±5.84	1.960	0.059
Height (m)	1.75±0.06	1.71±0.06	2.644	0.013
BMI (kg/m²)	22.14±1.59	21.75±2.15	0.759	0.454
Sport experience (years)	6.63±3.77	6.18±3.63	0.329	0.744
Training (days/week)	3.94±1.29	3.94±1.25	0.139	0.891
Lateral/medial hamstring injury	7/9			
Time since injuries (months)	9.38±5.45			
Time to return to play (weeks)	3.94±4.33			

HSI: hamstring strain injuries; CON: control.

Materials and methods

Participants

Sixteen male athletes with a history of HSI and 16 matched controls were recruited from track and field sports to a single-session cross-sectional study (Table I). The inclusion criteria for the injured group were athletes with a history of HSI over the past 3-24 months, but no pain or discomfort in the muscle on the testing date. The control group had no history of HSI and was recruited based on the matched sports and characteristics of the participants in the injured group. The exclusion criteria were hamstring avulsion, referred pain from other areas that were not related to hamstring strain, and severe lower limb injuries in the past six months. To ensure adequate statistical power to detect a significant difference between groups, the sample size was determined using the G*Power 3.1.9.7 program (G-Power, Brunsbüttel, Germany) based on pilot study results. The calculated effect size was 0.8, indicating a medium to large effect. A minimum of 28 participants was required in this study to achieve a statistical power of 80% with an alpha level of 0.05. Before commencing the jumping trials, the participants were informed of the testing procedure and provided written informed consent, which was approved by the Research Ethics Committee of the university hospital in accordance with the Helsinki Declaration (the approval number: A-ER-110-075).

Hamstring mechanical properties

The mechanical properties of the hamstring in the relaxed state were measured using a MyotonPRO Digital Palpation Device (Myoton, Tallinn, Estonia), whose measurements are in good agreement with those obtained from shear wave ultrasound elastography.[23] In the measurement process, the probe was set perpendicular to the skin of the muscle of interest. For each participant, the lateral and medial hamstring bellies were assessed in three trials at each testing

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point, and the mean value calculated from the lateral and medial hamstrings of the leg was computed for analysis. The same investigator performed all the measurements and analyses to maintain consistency and reduce the risk of inter-rater variability.[24] The intraclass correlation coefficients (ICC) indicated a moderate-to-high reliability of the measurements (tone: 0.929, stiffness: 0.938, elasticity: 0.652). The participants were instructed to refrain from exercise for 48 hours prior to testing to minimize the effects of training on the muscle mechanical properties. Three outcomes were extracted for each participant: 1) muscle tone (Hz) = f_{max}, where f is the Fast Fourier transform of the MyotonPRO signal spectrum; 2) dynamic stiffness (N/m) = $a_{max} \cdot m_{prob}/\Delta l$, where a_{max} is the maximum acceleration of the damped oscillation, m_{prob} is the mass of the measurement mechanism, and Δl is the maximum displacement of the tissue; and 3) elasticity = ln (a_1/a_3), where a_1 is the maximum displacement (*i.e.*, maximum tissue resistance) measured in mG and a_3 is the maximum displacement of the second oscillation period, which is associated with the recuperation of the stored residual mechanical energy in the tissue.

Hamstring range of motion

The hamstring flexibility was measured in passive straight leg raise tests (PSLR) and active knee extension tests (AKE) using a MicroFET3 dynamometer (Hoggan Health Industries, Salt Lake City, UT, USA) strapped to the lateral border of the tibia.[25, 26] The zero degree of the dynamometer was set at the starting position in each trial before the tested limb was raised to the end of the range of motion (RoM). Three PSLR and AKE trials were conducted for each limb. For each trial, the RoM was measured at the point of maximal tolerable stretch, and the maximum RoM values obtained from the three trials of each exercise were averaged to obtain an accurate representation of the hamstring flexibility.

Hamstring strength

The isometric hamstring strength was measured in the supine position using the MicroFET3 dynamometer for two different contraction angles.[27] In the first case, the dynamometer was placed 3 cm above the heel and against the treatment table with the hip and knee straight (0-0°). In the second case, the dynamometer was positioned against a chair that was adjusted such that the seating height matched the upper thigh length of the participant to achieve passive flexion of the hip and knee at 90° (90-90°). The participants performed isometric contraction of the hamstring muscle to the maximum voluntary contraction of the knee flexor muscle in three trials for each range, with 1-minute rest intervals between trials of the same range and 2-minute rest intervals between trials of different ranges to avoid fatigue. For each range, the highest muscle force was recorded, and the average value across the three trials was used to indicate the hamstring strength.

Jumping mechanics

Two jumping tasks, squat jump (SJ) and countermovement jump (CMJ), were performed to observe the jump-landing mechanics. The SJ commenced with the participants lowering themselves into a squat position on separate force platforms (Kistler Instrument Corp., Winterthur, Switzerland) placed between the legs by bending the hips and knees simultaneously to 90° of flexion. The participants then jumped upward with maximum effort by extending their hips, knees, and ankles to propel their bodies into the air and land on the same force platforms. The CMJ began with the participants standing on the force platforms with their feet placed shoulder-width apart. The participants then initiated a quick and fluid downward movement by rapidly flexing their hips and knees as deep as possible, followed by jumping up into the air and landing on the same force platforms. For each task, the force measurements obtained from three accepted jumps were used for further analysis. To prevent neuromuscular fatigue, 1-minute rest intervals were introduced between each jumping trial.

Data processing

The vertical ground reaction force (vGRF) was filtered using a low-pass Butterworth digital filter with a cutoff frequency of 100 Hz. The peak vGRF during takeoff was determined from the maximum value of the takeoff force and normalized to the individual's body weight in Newtons (N). The jumping height in both tasks was computed using the flight time method[28] to quantify the jumping performance of each individual. The peak vGRF and loading rate during landing were normalized by the individual's body weight and time to reach the first peak vGRF, respectively, to quantify the landing performance.[29] Data processing was performed using MATLAB (MathWorks, R2021a, Natick, MA, USA).

Statistical analysis

The data were summarized using the mean, standard deviation (SD), and 95% confidence interval (95% CI). The Shapiro-Wilk Test was used to assess the normality assumption of the outcome measures. The results indicated

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that all the outcomes were normally distributed. Independent *t*-tests were used to identify statistically significant differences in the characteristics of the participants and the jump height outcomes between the injured and control groups. One-Way ANOVA was conducted to compare the main outcomes (hamstring mechanical properties, hamstring flexibility, hamstring strength, and jump-landing mechanics) among the injured limb, non-injured limb, and matched-control limb. The Bonferroni *post-hoc* test was applied when the ANOVA results indicated significant differences among the group means. Cohen's *d* effect size was evaluated to quantify the magnitude of the statistical significance for each outcome. Additionally, Pearson's correlation coefficient was computed to examine the correlations between the hamstring mechanical properties and contractile functions and the jump-landing mechanics. The statistical analyses were performed using SPSS version 26 (IBM Corp., Armonk, NY, USA). The significance level was set at P<0.05.

Results

Hamstring mechanical properties and contractile function

There were significant differences in muscle tone ($F_{(2,45)}$=3.854, P=0.036) and stiffness ($F_{(2,45)}$=4.931, P=0.012). The injured group exhibited a higher muscle tone (P=0.042, *d*=0.892) and stiffness (P=0.010, *d*=1.102) in the injured leg compared to the matched controls. In addition, the HSI group showed a lower hamstring flexibility ($F_{(2,45)}$=7.789, P=0.001) of the injured leg than the control group in the AKE tests (P=0.002, *d*=1.279) and a reduced hamstring isometric strength ($F_{(2,45)}$=3.740, P=0.031) of the injured leg than the control group at hip and knee angles of 0°-0° (P=0.040, *d*=0.975). No obvious differences were observed between the hamstring mechanical properties, flexibility, and strength of the injured and non-injured legs (Table II).

Jump-landing performance

For the CMJ task, the control group can jump higher than the injured group with *t*(30)=-2.182, P=0.037, *d*=0.771 (control: 48.53±6.40 cm *vs.* injured: 43.71±6.10 cm). Differences in the measured outcomes between the injured leg and the matched control leg were observed for the peak vGRF at landing ($F_{(2,45)}$=5.858, P=0.005) and loading rate ($F_{(2,45)}$=6.580, P=0.003) (Table III). However, no difference in the kinetic outcomes was found between the injured and control legs in the SJ task or between the injured and non-injured legs in either task.

Associations among hamstring mechanical properties, contractile function, and jump-landing mechanics

The hamstring muscle tone was associated with the jump height in both the SJ task (*r*=0.312, P=0.031) and the CMJ task (*r*=0.290, P=0.045). The stiffness and elasticity were

TABLE II.—*Comparisons of hamstring properties between the injured and matched control groups.*

Parameters	INJ	Non-INJ	CON	*F*	P	*Post-hoc* tests	Cohen's *d* (INJ *vs.* non-INJ)	Cohen's *d* (INJ *vs.* CON)	Cohen's *d* (Non-INJ *vs.* CON)
Hamstring mechanical properties									
Tone (Hz)	19.20±1.34 (18.54-19.86)	18.94±1.23 (18.34-19.54)	18.07±1.19 (17.49-18.65)	3.854	0.036*	INJ *vs.* CON: P=0.042	0.202	0.892	0.719
Stiffness (N/m)	360.17±31.88 (344.55-375.79)	347.81±29.78 (333.22-362.40)	327.56±27.09 (314.29-340.83)	4.931	0.012*	INJ *vs.* CON: P=0.010	0.401	1.102	0.711
Elasticity	1.33±0.08 (1.29-1.37)	1.33±0.09 (1.29-1.37)	1.28±0.07 (1.25-1.31)	2.135	0.130		0.000	0.665	0.620
Hamstring flexibility (°)									
PSLR	78.00±7.00 (74.57-81.43)	77.21±6.41 (74.07-80.35)	81.83±7.50 (78.16-85.51)	0.964	0.389		0.118	0.528	0.662
AKE	54.56±9.48 (49.92-59.21)	55.98±9.32 (51.42-60.54)	64.23±4.97 (61.80-66.67)	7.789	0.001*	INJ *vs.* CON: P=0.002 Non-INJ *vs.* CON: P=0.008	0.151	1.279	1.106
Hamstring strength (N.)									
At 0-0°	207.99±29.13 (193.72-22.26)	214.30±41.46 193.99-234.62	241.59±39.07 (222.45-60.73)	3.740	0.031*	INJ *vs.* CON: P=0.040	0.176	0.975	0.677
At 90-90°	266.14±53.45 (239.95-92.33)	259.43±51.26 (234.31-84.55)	307.51±63.08 (276.60-38.42)	3.075	0.056		0.128	0.708	0.837

INJ: injured limb; Non-INJ: non-injured limb; CON: control group; PSLR: passive straight leg raise test; AKE: active knee extension test; Cohen's *d*: Cohen's *d* effect size.
*Statistically significant at P<0.05.

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TABLE III.—*Comparisons of squat jump and countermovement jump performance between the injured and matched control groups.*

Parameters	INJ	Non-INJ	CON	*F*	P	*Post-hoc* tests	Cohen's *d* (INJ *vs.* non-INJ)	Cohen's *d* (INJ *vs.* CON)	Cohen's *d* (Non-INJ *vs.* CON)
Peak vGRF at takeoff (%BW)									
SJ	1.23±0.14 (1.16-1.30)	1.20±0.10 (1.151-1.249)	1.21±0.13 (1.146-1.274)	0.185	0.832		0.247	0.148	0.086
CMJ	1.44±0.25 (1.32-1.56)	1.41±0.19 (1.307-1.493)	1.27±0.15 (1.197-1.343)	3.063	0.057		0.135	0.825	0.818
Peak vGRF at landing (%BW)									
SJ	2.57±1.15 (2.01-3.13)	2.65±1.21 (2.06-3.24)	2.10±0.63 (1.79-2.41)	1.299	0.283		0.068	0.507	0.570
CMJ	3.03±0.95 (2.57-3.50)	2.95±0.86 (2.53-3.37)	2.11±0.69 (1.77-2.45)	5.858	0.005*	INJ *vs.* CON: P=0.011 Non-INJ *vs.* CON: P=0.021	0.088	1.108	1.077
Loading rate (%BW·s^{-1})									
SJ	76.81±62.67 (46.10-107.52)	71.52±66.71 (38.83-104.21)	49.12±37.35 (30.82-67.42)	1.061	0.355		0.082	0.537	0.414
CMJ	80.98±54.46 (54.30-107.67)	70.83±41.04 (50.72-90.94)	31.77±16.56 (23.66-39.88)	6.580	0.003*	INJ *vs.* CON: P=0.004 Non-INJ *vs.* CON: P=0.027	0.210	1.223	1.248

INJ: injured limb; Non-INJ: non-injured limb; CON: control group; vGRF: vertical ground reaction force; SJ: squat jump; CMJ: countermovement jump; Cohen's *d*: Cohen's *d* effect size.
*Statistically significant at P<0.05.

TABLE IV.—*Associations between hamstring properties and jump-landing performance.*

Parameters	Squat jump				Countermovement jump			
	TO force	Jump height	LD force	LR	TO force	Jump height	LD force	LR
Hamstring mechanical properties								
Tone (Hz)	*r*=0.010 P=0.945	*r*=0.312* P=0.031*	*r*=-0.329* P=0.022*	*r*=-0.247 P=0.090	*r*=-0.090 P=0.541	*r*=0.290* P=0.045*	*r*=-0.153 P=0.298	*r*=0.102 P=0.492
Stiffness (N/m)	*r*=-0.006 P=0.970	*r*=0.199 P=0.175	*r*=-0.288* P=0.047*	*r*=-0.211 P=0.151	*r*=-0.075 P=0.615	*r*=0.204 P=0.164	*r*=-0.058 P=0.694	*r*=0.211 P=0.150
Elasticity	*r*=0.009 P=0.949	*r*=-0.008 P=0.959	*r*=0.121 P=0.414	*r*=0.175 P=0.234	*r*=0.084 P=0.570	*r*=-0.028 P=0.850	*r*=0.249 P=0.087	*r*=0.287* P=0.048*
Hamstring flexibility (°)								
PSLR	*r*=0.179 P=0.224	*r*=0.102 P=0.488	*r*=0.284 P=0.050	*r*=0.249 P=0.088	*r*=0.186 P=0.207	*r*=0.133 P=0.368	*r*=0.114 P=0.440	*r*=0.124 P=0.400
AKE	*r*=0.196 P=0.181	*r*=0.404** P=0.004*	*r*=-0.056 P=0.704	*r*=-0.031 P=0.836	*r*=0.043 P=0.769	*r*=0.416** P=0.003*	*r*=-0.205 P=0.162	*r*=-0.056 P=0.705
Hamstring strength (N)								
At 0°-0°	*r*=0.060 P=0.687	*r*=0.166 P=0.258	*r*=-0.495** P<0.001*	*r*=-0.453** P=0.001*	*r*=-0.374** P=0.009*	*r*=0.308* P=0.033*	*r*=-0.414** P=0.003*	*r*=-0.349* P=0.015*
At 90°-90°	*r*=0.127 P=0.391	*r*=0.338* P=0.019*	*r*=-0.318* P=0.027*	*r*=-0.362* P=0.012*	*r*=-0.155 P=0.292	*r*=0.280 P=0.054	*r*=-0.226 P=0.123	*r*=-0.212 P=0.149

PSLR: passive straight leg raise test; AKE: active knee extension test; TO: takeoff; LD: landing; LR: loading rate.
**Statistically correlation at the 0.01 level; *statistically significant correlation at the 0.05.

correlated with the landing force in the SJ task (*r*=-0.288, P=0.047) and the loading rate in the CMJ task (*r*=0.287, P=0.048). No correlation was observed between the PSLR and jumping mechanics in either task. However, the AKE showed a moderate correlation with the jump height in both tasks (*r*=0.404, P=0.004 for the SJ and *r*=0.416, P=0.003 for the CMJ). In addition, the hamstring strength at 0-0° was correlated with both the landing force (*r*=-0.495, P=0.000 for the SJ and *r*=-0.414, P=0.003 for the CMJ) and the loading rate (*r*=-0.453, P=0.001 for the SJ and *r*=-0.349, P=0.015 for the CMJ). Finally, the hamstring strength at 90-90° was correlated with both outcomes in the SJ task but was unrelated to the jump performance in the CMJ task (Table IV).

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Discussion

This study investigated alterations in the hamstring properties of athletes with HSI, together with the impact of these alterations on the jump-landing performance, compared to healthy controls. The results revealed that the individuals with HSI had higher muscle tone and stiffness than the control group, but lower muscle flexibility and strength. However, no significant differences were detected between the injured and non-injured legs. Changes in the mechanical properties of the hamstring due to prior HSI contributed to an impaired jumping and landing performance, with a mild-to-moderate correlation noted between the changes in the hamstring properties and the task outcomes. The results are thus consistent with the hypothesis that an impaired function of the hamstring after HSI leads to a poorer jump-landing performance compared with the non-injured limb or matched limb in healthy controls.

The results of our study revealed that both hamstring muscle tone and stiffness in the injured leg were higher than those in the control leg. Although no significant differences in muscle tone and stiffness were detected between the injured and non-injured legs, the injured leg tends to exhibit higher tone and stiffness than the non-injured leg. Previous studies have shown that the mechanical properties of the hamstring are remodeled after strain to meet the metabolic and functional demands of the muscle.[30] Interestingly, the hamstring mechanical properties observed in the present study were better than those reported in previous cohort studies that also used the MyotonPro device to measure the outcome. For example, the hamstring stiffness in athletes with HSI was documented as 330±33 N/m at 26 weeks in Reurink's study[9] and as 343.13±13.38 N/m at 29.25 months postinjury onset in Kawai *et al.*'s study.[10] Moreover, the muscle tone was reported as 17.5±1.6 Hz at 26-week post-HSI assessment in Reurink's study.[9] In the current study, the hamstring tone and stiffness of the injured leg were reported as 19.20±1.34 Hz and 360.17±13.38 N/m, respectively, at an average of 9 months after injuries. This variation across studies may be attributed to several factors, including the characteristics of the participants, the initial injury severity and location, the testing time point, the implemented rehabilitation programs, and the exercise intensity in the recovery phase and after return to sports.[12] Overall, however, the present results provide further evidence of persistent long-term muscle remodeling after HSI.

The increased mechanical properties of the hamstring following HSI could be explained by the fact that visible structural damage caused by acute injury is often accompanied by swelling and fluid accumulation, which eventually leads to a redistribution of the collagen and vascular infrastructures[7] and the development of scar tissue.[8] In particular, prolonged fibroblast activity during the healing process can result in the continuous production of collagen and other extracellular matrix (ECM) components. This in turn promotes an excessive production of collagen, which causes abnormal cross-linking and a less well-organized and random orientation of the collagen fibers that creates a more rigid and less elastic structure.[31] The replacement scar tissue, with differences in its composition and structure primarily attributed to the altered arrangement of the collagen fibers and the presence of other ECM components, disrupts the normal architecture of the muscle fibers and leads to a greater muscle tone or stiffness, together with a lower elasticity of the tissue sliding system.[31] The development of scar tissue after HSI was identified as early as 6 weeks in Connell *et al.*' study[11] and was found to persist for up to 23 months in Silder *et al.*'s study.[12] Since the athletes in the present study had an average HSI history of 9 months, it is possible that all the injured muscles were prone to scar tissue formation, leading to a higher tone and/or stiffness in the injured limb than in the non-injured limb or matched-control limb.

In terms of the contractile function, the present study found a statistically significant difference in the hamstring flexibility, measured by AKE, and the hamstring strength at hip and knee flexions of 0°, between the injured and control limbs. However, no such differences were observed between the injured and non-injured limbs. The results thus suggest a link between the changes in the mechanical properties of the hamstring and the muscle contractile function deficits. A higher muscle tone and stiffness, along with an impaired elasticity caused by scar tissue, can disrupt the normal architecture of the muscle fibers, affecting the contractile function of the elastic tissues by reducing the muscle flexibility and optimal length for active force generation. Since the sliding system underpins the resilience of fascial tissue structures,[32] a weakness of the hamstring, attributed to scarring in the injured muscle, can impair the ability of the hamstring to generate force during activities that require rapid and powerful movements. In particular, hamstring weakness disrupts normal neuromuscular control of the muscle and thus negatively affects the precision and timing of muscle activation, recruitment, and coordination during contraction.[12] Silder *et al.*[12] suggested that the loss of muscle strength after HSI may be the result of muscle atrophy, a condition where muscle mass loss persists even after the initial healing phase for periods as

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long as 5 to 26 months. A recent study by Avrillon *et al.*[33] confirmed that selective atrophy of the injured hamstring decreases the contribution of the total knee flexor torque, potentially impacting the stability, function, and overall biomechanics of the hip and knee joints. Hence, it appears that a combination of factors in the injured muscle may contribute to deficits in the muscle contractile function.

Jumping and landing tasks are common in sports activities and training and are often used to assess readiness for return to play or the risk of injury recurrence after hamstring strains.[34] To gain further insights into how changes in the hamstring properties impact the contractile function of the muscle during functional tasks, this study compared the jumping mechanics outcomes among different limbs and populations. The results revealed that long-term alterations in the hamstring properties affected the jumping performance in HSI athletes, with mild-to-moderate correlations between the hamstring properties and the jumping task outcomes. In general, the results suggest that the mechanical properties of the muscles around the hip and knee, especially the muscle tone and stiffness, play a vital role in regulating the control and stability of the hip and knee joints during takeoff and landing. Altered hamstring properties compromise the biomechanical stability required for effective and controlled movements during takeoff and landing. For instance, a higher tone and stiffness of the injured hamstring impair the contractile function and activation level of the contracted muscle. Furthermore, deficits in the hamstring strength and flexibility lead to insufficient power and a suboptimal degree of movement of the injured limb in the push-off stage during jumping, which lowers the push-off force and degrades the jump performance. Similarly, in the landing phase, a greater muscle tone or stiffness, together with a lower strength and flexibility, hinders the ability of the muscle to absorb shock and dissipate the landing force. Moreover, changes in the hamstring properties also affect the neuromuscular control ability of the muscle, degrading the coordination and timing of muscle coupling, such as that between the agonistic and antagonistic muscles. The resulting impaired control over movements of the hip and knee joints increases the risk of injury during landing.[35] Interestingly, in the present study, these phenomena were observed only in the countermovement jump. This finding is reasonable since countermovement jumps mainly rely on the elastic tissue component of the coupling muscle and the reflex mechanism to perform the task through the stretch-shortening cycle. The combined stimulation of the prestretch mechanism and the Golgi tendon reflex enhances the neural drive and muscle activation, leading to more forceful and explosive muscle contraction during the jump task.[36, 37] In contrast, the squat jump commences with isometric contraction of the hamstring and involves no prestretch mechanism. Consequently, the two jumping tasks entail different contributions from the hamstrings and thus elicit different jump task outcomes.

The present findings showed that alterations in the hamstring mechanical properties and deficits in the contractile function impacted the jump-landing mechanics in individuals with chronic HSI. These findings have significant importance for clinical practice in that they suggest the need to evaluate the hamstring mechanical properties in athletes with prior HSI since these properties may impact their dynamic and explosive movement performance. It is thus recommended that rehabilitation programs for HSI should prioritize addressing the tissue property factors of the muscle through a comprehensive approach encompassing stretching, strengthening exercises, neuromuscular retraining, and a gradual return to sports activity, even when the return to play criteria are satisfied and athletes can return to play without persistent symptoms in the injured leg.[7, 38] Such an approach would likely aid in improving muscle function and sports performance, while simultaneously reducing the risk of reinjury, which is a common phenomenon in athletes engaged in a wide range of sports activities.

Limitations of the study

This study has several limitations. Firstly, all the participants were male. Since a previous study indicated differences in the hamstring properties across sexes,[39] generalization of the present findings to female athletes should be performed with caution. Secondly, the participants in the study were recruited based on their HSI injury history without diagnostic confirmation by medical imaging, such as MRI or ultrasound. Thus, future studies should consider investigating the injury history using imaging techniques to obtain additional information and a deeper theoretical framework for the study. Finally, the MyotonPRO device was used to assess the passive mechanical properties of the soft hamstring tissue. However, the contractile function of the hamstring muscle is likely affected by both its passive and active mechanical properties. Consequently, future studies are recommended to scrutinize the active mechanical properties of the hamstring to provide a more comprehensive understanding of the impact of both the active and passive mechanical properties of the injured muscle on the outcomes of various tasks.

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Conclusions

Athletes with a history of hamstring strain exhibited alterations in the hamstring mechanical properties and contractile function of the injured limb compared to matched controls. Notably changes in the hamstring mechanical properties and contractile function of the injured leg adversely impacted the jump performance and landing strategy compared with the control group. Finally, the hamstring mechanical properties and contractile function showed mild-to-moderate correlations with the jumping task outcomes. Overall, the findings suggest the need to assess the hamstring mechanical properties and contractile function in individuals with chronic HSI since they impact movement performance and may increase the injury risk. In addition, rehabilitation programs aimed at restoring hamstring function should focus primarily on comprehensive rehabilitation strategies and/or targeted training regimens.

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Conflicts of interest
The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Authors' contributions
Both authors contributed equally to the manuscript, Amornthep Jankaew and Cheng-Feng Lin have given substantial contributions to study conception and design, and manuscript writing, Amornthep Jankaew to data collection, analysis, and interpretation, Cheng-Feng Lin to manuscript critical revision. Both authors read and approved the final version of the manuscript.

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